

Andrés MARTÍNEZ

Research Professor in Data Science /
PhD Candidate in Management



PROFILE

An expert in Finance and Economics, with a Master of Science in Stochastic Engineering from the **Munich University of Applied Sciences**, Germany. Presently serving as a full-time research professor in the Department of Mathematics at **Universidad Externado**, and concurrently engaged in a PhD program in Management at **Universidad de los Andes**. The research focuses on pricing techniques, employing Bayesian estimation within stochastic and data science models, with an emphasis on price prediction and risk assessment in financial instruments.

CONTACT DETAILS

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✉ Bogotá, Colombia

PERSONAL INFO

Date of birth: 18/12/1984
Nationality: Colombian

AREAS OF SPECIALIZATION

- Data Science
- Quantitative Finance
- Bayesian Statistics

LANGUAGES

Spanish (Native)
German (B2)
English (B2)

PROGRAMMING

R and Python

EXPERIENCE

FULL-TIME PROFESSOR (DATA SCIENCE) at *Mathematics Dept. Universidad Externado (Colombia)* **2017–Current**

◊ Lead the data science research group and coordinate data science projects. Teach Bachelor's and Master's level courses, including Introduction to Data Science, Stochastic Calculus, Bayesian Statistics, Data Mining, and Data Science for Smart Cities.

VISITING PHD STUDENT (ECONOMICS) at *Ca' Foscari, Venice (Italy)* **2022–2022**

◊ Under the supervision of Professor Roberto Cassarin, I have contributed to the paper entitled "Cryptoassets Market Structure: A Bayesian Nonparametric Approach."

FULL-TIME PROFESSOR (FINANCE) at *Universidad de La Salle (Colombia)* **2019–2020**

◊ Taught Risk Management, Derivatives, and Econometrics. Directed the Financial Laboratory.

EDUCATION

PHD IN MANAGEMENT (CANDIDATE) *Universidad de los Andes (Bogotá, Colombia)* **2021–2025**

◊ Research focuses on Bayesian estimation and stochastic/data science models.

M.SC. IN STOCHASTIC ENGINEERING *Hochschule München (Munich, Germany)* **2012–2014**

◊ Advanced probability, econometric modelling, time-series forecasting.

B.A. IN ECONOMICS *Universidad Sergio Arboleda (Bogotá, Colombia)* **2006–2010**

◊ Focus on financial economics, quantitative methods.

TALKS

2024: *Structure of the Crypto-Asset Market: A Non-Parametric Bayesian Approach.* **Bayes Plurinacional, Salta (Argentina)**

2020: *Volatility Models for Currencies Returns in Latin America.* **Congreso Redradar, Buenos Aires (Argentina)**

◊ An application of the Heston–Nandi model.

2018: *Allocation of Fixed Income Instruments using a Multivariate Model in the Colombian Market.*

IX Congreso de Investigación Financiera, Monterrey (México)

PUBLICATIONS

2021: *Relevancia del patrón de persistencia de Hurst en la gestión de portafolios de renta variable.*

Revista de Métodos Cuantitativos para la Economía y la Empresa, 32, 66–82.

2016: *Pronóstico de un Título de Renta Fija en Colombia.* ISSN: 2448–6795.

ACADEMIC NOTES

- Member of Italian Econometric Association (since 2024)
- ORCID: 0000-0003-0792-0144